

Course: Mathematics of Music	CIE-3 - SCHEME	Maximum Marks: 10+50
Course Code: MA375TGR	Seventh Semester 2025-26	Time: 1:20 PM to 3:20 PM
	UG	Date: 14/01/2026

QUIZ SECTION ($5 \times 2 = 10$ Marks)

Q.No.	Questions	M
1.	Āvartana: One complete cycle of tāla, begins/ends at sama.	1
	Sama: First beat of cycle, primary reference point for compositions.	1
2.	Formula: $3P + 2G = T$	1
	P = pattern length, G = gap between repetitions, T = total duration.	1
3.	Fractal: Object exhibiting self-similarity at different scales.	1
	Example: Koch Snowflake / Sierpinski Triangle / Cantor Set.	1
4.	Critical Band: Frequency bandwidth where sounds interact perceptually.	1
	Significance: Within CB $\rightarrow$ dissonance; separated by CB $\rightarrow$ consonance.	1
5.	Rūpaka Tāla: O I (Dhrutam + Laghu). Miśra jāti: Laghu = 7.	1
	Total = $2 + 7 = 9$ akṣaras.	1

TEST SECTION ($5 \times 10 = 50$ Marks)

Q.No.	Questions	M
1a	Mātrā: Smallest time unit, indivisible pulse.	1
	Akṣara: Fundamental beat unit, contains mātrās.	1
	Āvartana: Complete cycle, begins/ends at sama.	1
	Jāti sets laghu length $j \in \{3,4,5,7,9\}$.	1
	Nadai sets mātrās per akṣara. Formula: $N_{\text{mātrā}} = N_{\text{akṣara}} \times \text{nadai}$.	1
1b	Structure I O I I with Khaṇḍa jāti (laghu = 5):	1
	Akṣaras = $5 + 2 + 5 + 5 = 17$ akṣaras.	2
	Miśra nadai (7 mātrās per akṣara):	1
	Total = $17 \times 7 = 119$ mātrās.	1
2a	Given: $T = 64$ mātrās. Formula: $3P + 2G = 64$.	1
	Choose $P = 10$: $3(10) + 2G = 64 \rightarrow 2G = 34 \rightarrow G = 17$.	2
	Verification: $10 + 17 + 10 + 17 + 10 = 64$ mātrās.	1
	Ends at sama since $64 = 2 \times 32$ (2 cycles of Ādi Tāla).	1
2b	Korvai: Pūrvāṅga = 12, Uttarāṅga = 8 mātrās.	1
	One statement = $12 + 8 = 20$ mātrās. Total = $3 \times 20 = 60$ mātrās.	1
	Rūpaka (Chatusra jāti): O I = $2 + 4 = 6$ akṣaras.	1
	With Chatusra nadai: $6 \times 4 = 24$ mātrās/cycle. 2 cycles = 48.	1
	$60 \neq 48 \rightarrow$ Does NOT fit. Overshoots by 12 mātrās.	1
3a	1/f Noise (Voss & Clarke, 1978): $P(f) \propto 1/f^\alpha$, $\alpha \approx 1$.	1

	Found in Bach, Beatles, radio — consistent spectral signature.	1
	Relation to entropy/redundancy:	1
	White ($\alpha=0$): chaotic, high entropy. Brown ($\alpha=2$): predictable.	1
	Pink ($\alpha=1$): optimal balance — musical, engaging.	1
3b	Entropy: $H = -\sum P(x_i) \log_2 P(x_i)$ $H = -[0.4(-1.322) + 0.3(-1.737) + 0.2(-2.322) + 0.1(-3.322)]$ $H = 0.529 + 0.521 + 0.464 + 0.332 = 1.846$ bits. $H_{\text{max}} = \log_2(4) = 2$ bits. $R = 1 - 1.846/2 = 0.077$ (7.7%). Interpretation: Low redundancy → unpredictable, engaging melody.	1 1 1 1 1
4	(i) Zwicker's Formula: $BW_c(f) = 25 + 75[1 + 1.4(f/1000)^2]^{0.69}$ At 520 Hz: $BW_c = 25 + 75[1 + 1.4(0.52)^2]^{0.69}$ $= 25 + 75[1.3786]^{0.69} = 25 + 93.4 \approx 118$ Hz. (ii) Minor third ratio: $2^{(3/12)} = 1.1892$. $f_2 = 520 \times 1.1892 \approx 618$ Hz. Difference = $618 - 520 = 98$ Hz. % of CB = $98/118 \times 100 = 83\%$. Since $83\% > 50\%$, approaching separate bands → CONSONANT. Roughness relationship: 5-50% CB → max dissonance (beats).	1 1 2 1 1 1 1 1 1
5a	First-order Markov: $P(X_n = j X_{n-1} = i) = p_{ij}$. Future state depends only on current state, not history. Transition Matrix (step=0.5, stay=0.3, skip=0.2): $C \rightarrow [0.3, 0.5, 0.2]$, $D \rightarrow [0.25, 0.3, 0.45]$, $E \rightarrow [0.2, 0.5, 0.3]$ Row sums: $1.0 + 1.0 + 1.0 \rightarrow$ Row-stochastic verified.	1 1 1 2
5b	Missing Fundamental: Perceive pitch not physically present. Brain reconstructs via pattern recognition of harmonic series. Given harmonics: 200, 300, 400, 500 Hz. $= 2f_0, 3f_0, 4f_0, 5f_0$ where $f_0 = 100$ Hz (GCD). Perceived pitch = 100 Hz (cognitive, not mechanical).	1 1 1 1 1